```
%L2.b. Controllability, Observability and Pole placement

% Problem 1
% To check the controllability and observability of the control system

% State space matrices
A=[0 1 0;0 0 1;-6 -11 -6];
B=[0;0;1];
C=[4 5 1];
% Observability matrix
Ob=obsv(A,C);
% Order of system
no=size(A,1);
% Check observability
if rank(Ob)==no
    disp('System is observable')
else
    disp('System is not observable')
end
```

```
System is not observable
```

```

% Controllability matrix
CO=ctrb(A,B);

% Check controllability
if rank(CO)==no
    disp('System is controllable')
else
    disp('System is not controllable')
end
```

```
System is controllable
```

```
% Manual controllability matrix
CO1=[B A*B A^2*B];
% Rank of controllability matrix
rank(CO1)
```

```
ans =
3
```

```
% Manual observability matrix
OB1=[C;C*A;C*A^2];
% Rank of observability matrix
rank(OB1)
```

```
ans =
2
```

```
%L2.b. Controllability, Observability and Pole placement
```

```
% Problem 2
% Pole Placement by Ackermann's Formula

% State space matrices
A=[0 1 0;0 0 1;-6 -11 -6];
B=[0;0;1];
C=[4 5 1];
% Open loop poles
eig(A)
```

```
ans = 3×1
   -1.0000
   -2.0000
   -3.0000
```

```
% Controllability matrix
CO=[B A*B A^2*B];
% Check controllability
nc=size(A,1);
if rank(CO)==nc
    disp('System is controllable and pole placement is possible')
else
    disp('System is not controllable and pole placement is not possible')
end
```

```
System is controllable and pole placement is possible
```

```
% Desired closed loop poles in matrix form
P=[-8.4563 0 0;
   0 -0.7718+0.5919i 0;
   0 0 -0.7718-0.5919i];
% Desired closed loop poles in vector form
P1=[-8.4563;
    -0.7718+0.5919i;
    -0.7718-0.5919i];
% Desired characteristic polynomial coefficients
poly(P)
```

```
ans = 1×4
    1.0000    9.9999   13.9992    7.9998
```

```
% Desired characteristic polynomial matrix
Phi=polyvalm(poly(P),A);
% Ackermann formula
K0=[0 0 1]*inv(CO)*Phi
```

```
K0 = 1×3
    1.9998    2.9992    3.9999
```

```
% Closed loop system matrix
Acl0=A-B*K0;
% Closed loop poles
eig(Acl0)
```

```
ans = 3×1 complex
  -8.4563 + 0.0000i
  -0.7718 + 0.5919i
  -0.7718 - 0.5919i
```

```
% Verification using acker
Kacker1=acker(A,B,P1);
Acl1=A-B*Kacker1;
eig(Acl1)
```

```
ans = 3×1 complex
  -8.4563 + 0.0000i
  -0.7718 + 0.5919i
  -0.7718 - 0.5919i
```

```
% Verification using place
Kacker2=place(A,B,P1);
Acl2=A-B*Kacker2;
eig(Acl2)
```

```
ans = 3×1 complex
  -8.4563 + 0.0000i
  -0.7718 + 0.5919i
  -0.7718 - 0.5919i
```

```
%L2.b. Controllability, Observability and Pole placement

% Exercise 1(a)
% Pole placement by controllable canonical form transformation method

% State space matrices
A=[0 1 0;0 0 1;-6 -11 -6];
B=[0;0;1];
C=[4 5 1];
% Open loop poles
eig(A)
```

```
ans = 3×1
   -1.0000
   -2.0000
   -3.0000
```

```
% System order
no=size(A,1);
% Controllability matrix
CO=ctrb(A,B);
% Rank of controllability matrix
rank(CO)
```

```
ans =
3
```

```
% Check controllability
if rank(CO)==no
    disp('System is controllable and arbitrary pole placement can be done')
```

```
else
    disp('System is not controllable and arbitrary pole placement is not possible')
end
```

```
System is controllable and arbitrary pole placement can be done
```

```
% Open loop characteristic polynomial
ja=poly(A);
% Extraction of coefficients
a1=ja(2);
a2=ja(3);
a3=ja(4);
% Weighting matrix
w=[a2 a1 1;
   a1 1 0;
   1 0 0];
% Transformation matrix
t=CO*w;
% Desired closed loop pole locations
j=[-1+2i 0 0;
   0 -1-2i 0;
   0 0 -6];
% Desired characteristic polynomial
jj=poly(j);
% Extraction of desired coefficients
alpha1=jj(2);
alpha2=jj(3);
alpha3=jj(4);
% Feedback gain matrix
k=[alpha3-a3 alpha2-a2 alpha1-a1]*inv(t)
```

```
k = 1×3
   24.0000    6.0000    2.0000
```

```
% Closed loop system matrix
Acl=A-B*k;
% Closed loop poles
eig(Acl)
```

```
ans = 3×1 complex
  -1.0000 + 2.0000i
  -1.0000 - 2.0000i
  -6.0000 + 0.0000i
```

```
%L2.b. Controllability, Observability and Pole placement

% Exercise 1(b)
% Pole placement by controllable canonical transformation method 2

% State space matrices
A=[0 1 0;0 0 1;-6 -11 -6];
B=[0;0;1];
```

```
C=[4 5 1];

% Open loop poles
eig(A)
```

```
ans = 3×1
   -1.0000
   -2.0000
   -3.0000
```

```
% System order
no=size(A,1);
% Controllability matrix
CO=ctrb(A,B);
% Rank of controllability matrix
rank(CO)
```

```
ans =
3
```

```
% Check controllability
if rank(CO)==no
    disp('System is controllable and arbitrary pole placement can be done')
else
    disp('System is not controllable and arbitrary pole placement is not possible')
end
```

```
System is controllable and arbitrary pole placement can be done
```

```
% Open loop characteristic polynomial
ja=poly(A);
% Extraction of characteristic polynomial coefficients
a1=ja(2);
a2=ja(3);
a3=ja(4);
% Last row of inverse controllability matrix
p1=inv(CO);
p1=p1(3,:);
% Transformation matrix
V=[p1;
   p1*A;
   p1*A*A];
% Controllable canonical form transformation
AC=V*A*inv(V)
```

```
AC = 3×3
    0.0000    1.0000    0.0000
         0    0.0000    1.0000
   -6.0000  -11.0000   -6.0000
```

```
BC=V*B
```

```
BC = 3×1
    0.0000
   -0.0000
```

```
    1.0000
```

```
% Last row of controllable canonical form matrix
alpha=AC(3,:)
```

```
alpha = 1×3
   -6.0000  -11.0000   -6.0000
```

```
% Desired closed loop poles
P=[-1+2i -1-2i -6];
% Desired characteristic polynomial
lambda=poly(P);
% Desired coefficients
lambda=lambda(2:4);
% State feedback gain matrix in CCF
K1=[alpha(1)+lambda(3) alpha(2)+lambda(2) alpha(3)+lambda(1)]
```

```
K1 = 1×3
   24.0000    6.0000    2.0000
```

```
% State feedback gain matrix for original system
K=K1*V
```

```
K = 1×3
   24.0000    6.0000    2.0000
```

```
% Closed loop poles
eig(A-B*K)
```

```
ans = 3×1 complex
  -1.0000 + 2.0000i
  -1.0000 - 2.0000i
  -6.0000 + 0.0000i
```

```
% Verification in controllable canonical form
eig(AC-BC*K1)
```

```
ans = 3×1 complex
  -1.0000 + 2.0000i
  -1.0000 - 2.0000i
  -6.0000 + 0.0000i
```

```
% Ackermann formula
Kacker=[0 0 1]*inv(CO)*(A*A*A+lambda(1)*A*A+lambda(2)*A+lambda(3)*eye(3))
```

```
Kacker = 1×3
   24.0000    6.0000    2.0000
```

```
% Verification using acker command
Kacker1=acker(A,B,P)
```

```
Kacker1 = 1×3
    24     6     2
```

```
%L2.b. Controllability, Observability and Pole placement
```

```
% Exercise 2
% Pole placement using place command
% Find the control gain F such that the closed loop poles are
% at [-1+2i -1-2i -6]
% Verify by finding the eigenvalues of A-BF

% State space matrices
A=[-1 0 0;
   1 -2 0;
   2 1 -3];
B=[10;
   1;
   0];
% Desired closed loop poles
P=[-1+2i -1-2i -6];
% Feedback gain matrix
F=place(A,B,P)
```

```
F = 1×3
   -0.2222    4.2222   -2.0000
```

```
% Closed loop system matrix
Acl=A-B*F
```

```
Acl = 3×3
    1.2222  -42.2222   20.0000
    1.2222   -6.2222    2.0000
    2.0000    1.0000   -3.0000
```

```
% Verification of closed loop poles
eig(Acl)
```

```
ans = 3×1 complex
  -1.0000 + 2.0000i
  -1.0000 - 2.0000i
  -6.0000 + 0.0000i
```

```
%L2.b. Controllability, Observability and Pole placement

% Exercise 3
% Obtain the state models in Jordan canonical form
% Plot the pole-zero map
% Determine controllability and observability
% Place the poles at -4, -8, -10

% State space matrices
A=[-2 -2.5 -0.5;
    1 0 0;
    0 1 0];
B=[1;
   0;
   0];
C=[1 4 3.5];
```

```
D=0;
% Jordan canonical form
[P,J]=jordan(A)
```

```
P = 3×3 complex
   0.0580 + 0.0000i  -0.5290 + 2.0079i  -0.5290 - 2.0079i
  -0.2408 + 0.0000i  -0.8796 - 1.1414i  -0.8796 + 1.1414i
   1.0000 + 0.0000i   1.0000 + 0.0000i   1.0000 + 0.0000i
J = 3×3 complex
  -0.2408 + 0.0000i   0.0000 + 0.0000i   0.0000 + 0.0000i
   0.0000 + 0.0000i  -0.8796 - 1.1414i   0.0000 + 0.0000i
   0.0000 + 0.0000i   0.0000 + 0.0000i  -0.8796 + 1.1414i
```

```
% Transformed state matrices
BJ=inv(P)*B
```

```
BJ = 3×1 complex
   0.5845 + 0.0000i
  -0.2923 - 0.1636i
  -0.2923 + 0.1636i
```

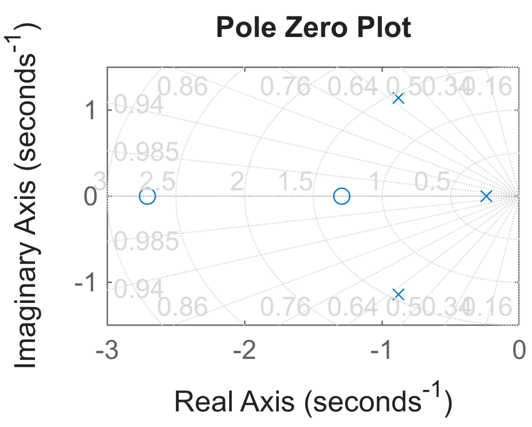
```
CJ=C*P
```

```
CJ = 1×3 complex
   2.5948 + 0.0000i  -0.5474 - 2.5575i  -0.5474 + 2.5575i
```

```
DJ=D
```

```
DJ =
0
```

```
% Pole-zero plot
[num,den]=ss2tf(A,B,C,D);
G=tf(num,den);
figure(1)
pzmap(G)
grid on
title('Pole Zero Plot')
```




```
% Controllability matrix
CO=ctrb(A,B);
% Rank of controllability matrix
```

```
rank(CO)
```

```
ans =
3
```

```
% Check controllability
if rank(CO)==size(A,1)
    disp('System is controllable')
else
    disp('System is not controllable')
end
```

```
System is controllable
```

```
% Observability matrix
OB=obsv(A,C);
% Rank of observability matrix
rank(OB)
```

```
ans =
3
```

```
% Check observability
if rank(OB)==size(A,1)
    disp('System is observable')
else
    disp('System is not observable')
end
```

```
System is observable
```

```
% Desired closed loop poles
Pdes=[-4 -8 -10];
% State feedback gain using place command
K=place(A,B,Pdes)
```

```
K = 1×3
   20.0000  149.5000  319.5000
```

```
% Closed loop system matrix
Acl=A-B*K;
% Verification of closed loop poles
eig(Acl)
```

```
ans = 3×1
  -10.0000
   -8.0000
   -4.0000
```

```
%L2.b. Controllability, Observability and Pole placement

% Exercise 4
% Place the poles of the system
% A=[-0.1 5 0.1;-5 -0.1 5;0 0 -10]
```

```
% B=[0;0;10]
% at (i) -1+j5, -1-j5, -10
% (ii) -5, -60, -70
% Find and compare the norms of F

% State space matrices
A=[-0.1 5 0.1;
   -5 -0.1 5;
    0 0 -10];
B=[0;
   0;
   10];
% Desired poles - Case 1
P1=[-1+5i -1-5i -10];
% Feedback gain matrix
F1=place(A,B,P1)
```

```
F1 = 1×3
   -0.1404    0.3754    0.1800
```

```
% Closed loop poles
eig(A-B*F1)
```

```
ans = 3×1 complex
  -1.0000 + 5.0000i
  -1.0000 - 5.0000i
 -10.0000 + 0.0000i
```

```
% Norm of F1
normF1=norm(F1)
```

```
normF1 =
0.4394
```

```
% Desired poles - Case 2
P2=[-5 -60 -70];
% Feedback gain matrix
F2=place(A,B,P2)
```

```
F2 = 1×3
   70.4864   94.5509   12.4800
```

```
% Closed loop poles
eig(A-B*F2)
```

```
ans = 3×1
   -5.0000
  -60.0000
  -70.0000
```

```
% Norm of F2
normF2=norm(F2)
```

```
normF2 =
118.5915
```

```
% Comparison of norms
disp('Norm of F for Case 1')
```

```
Norm of F for Case 1
```

```
disp(normF1)
```

```
    0.4394
```

```
disp('Norm of F for Case 2')
```

```
Norm of F for Case 2
```

```
disp(normF2)
```

```
  118.5915
```

```
%L2.b. Controllability, Observability and Pole placement

% Exercise 5
% A SISO system represented by
% A=[-1 0 0;0 -2 -2;-1 0 -3]
% Consider two cases
% B1=[1;1;1]
% B2=[1;1;0]
% i) Check controllability
% ii) Find eigenvalues
% iii) Find transformation matrix V that transforms the system
%      to controllable canonical form
% iv) Determine state feedback matrix K required to assign
%      eigenvalues at [-2 -4 -6]

% State matrix
A=[-1 0 0;
    0 -2 -2;
   -1 0 -3];
% Case 1
B1=[1;
    1;
    1];
% Eigenvalues
eig(A)
```

```
ans = 3×1
    -2
    -3
    -1
```

```
% Controllability matrix
CO1=ctrb(A,B1);
% Rank of controllability matrix
rank(CO1)
```

```
ans =
3
```

```
% Check controllability
if rank(CO1)==size(A,1)
    disp('Case 1: System is controllable')
else
    disp('Case 1: System is not controllable')
end
```

```
Case 1: System is controllable
```

```
% Transformation matrix
p1=inv(CO1);
p1=p1(3,:);
V1=[p1;
    p1*A;
    p1*A*A]
```

```
V1 = 3×3
         0    0.3333   -0.3333
    0.3333   -0.6667    0.3333
   -0.6667    1.3333    0.3333
```

```
% Desired poles
P=[-2 -4 -6];
% State feedback gain
K1=place(A,B1,P)
```

```
K1 = 1×3
    7.0000         0   -1.0000
```

```
% Verification
eig(A-B1*K1)
```

```
ans = 3×1
   -2.0000
   -6.0000
   -4.0000
```

```
% Case 2
B2=[1;
    1;
    0];
% Controllability matrix
CO2=ctrb(A,B2);
% Rank of controllability matrix
rank(CO2)
```

```
ans =
3
```

```
% Check controllability
if rank(CO2)==size(A,1)
    disp('Case 2: System is controllable')
```

```
else
    disp('Case 2: System is not controllable')
end
```

```
Case 2: System is controllable
```

```
% Transformation matrix
p2=inv(CO2);
p2=p2(3,:);
V2=[p2;
    p2*A;
    p2*A*A]
```

```
V2 = 3×3
    -1     1    -1
     2    -2     1
    -3     4     1
```

```
% State feedback gain
K2=place(A,B2,P)
```

```
K2 = 1×3
    6.0000         0   -3.0000
```

```
% Verification
eig(A-B2*K2)
```

```
ans = 3×1
   -2.0000
   -6.0000
   -4.0000
```